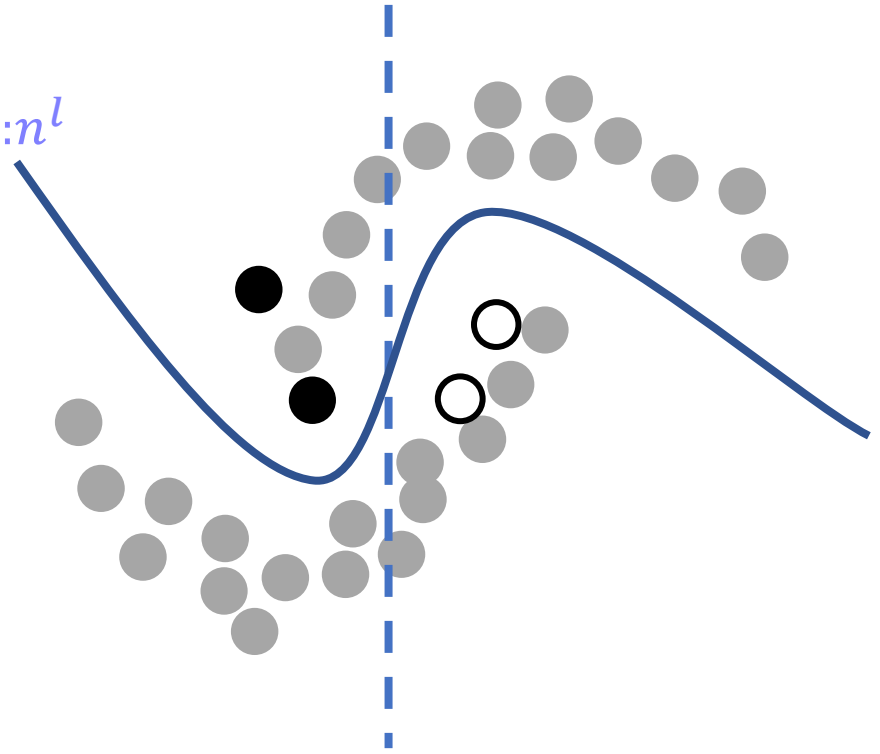
Semi-supervised Learning



General Formulation

- Labelled Dataset: $\mathcal{D}^l = \{(\mathbf{x}_i^l, y_i^l)\}_{i=1:n^l}$
- Unlabelled Dataset: $\mathcal{D}^u = \{\mathbf{x}_i^u\}_{i=1:n^u}$
- $n^u \gg n^l$

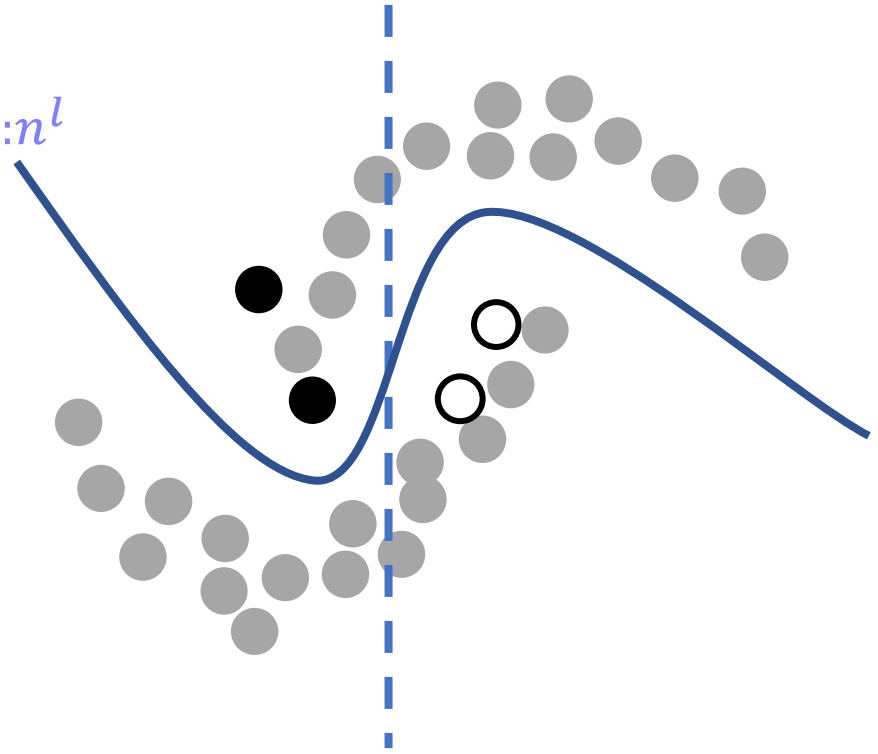


- $\boldsymbol{\theta}^* = \text{argmin}_{\boldsymbol{\theta}}\, \mathcal{L}(\mathcal{D}^l)$
  $= \text{argmin}_{\boldsymbol{\theta}}\, \sum_{i=1}^{n} l(f_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i) + \Omega(\boldsymbol{\theta})$

- $\boldsymbol{\theta}^* = \text{argmin}_{\boldsymbol{\theta}}\, \mathcal{L}(\mathcal{D}^l) + \mathcal{L}^u(\mathcal{D}^l, \mathcal{D}^u)$

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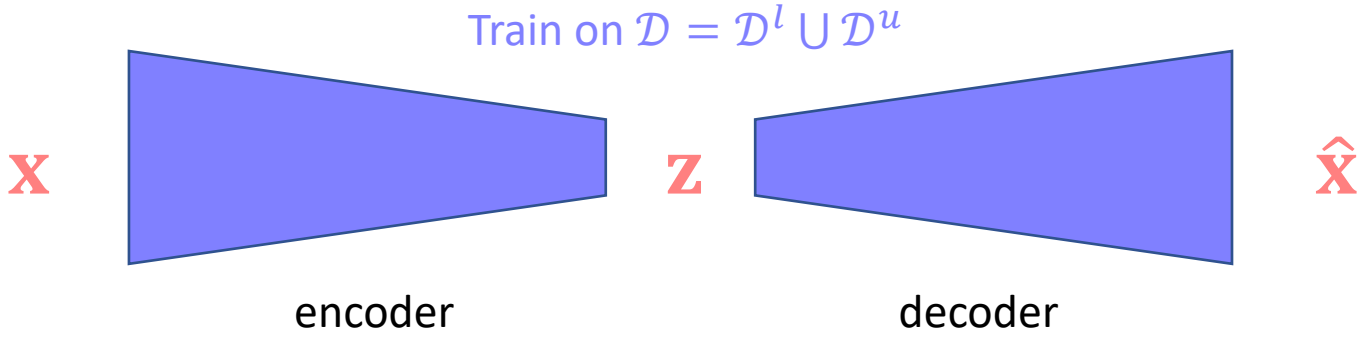


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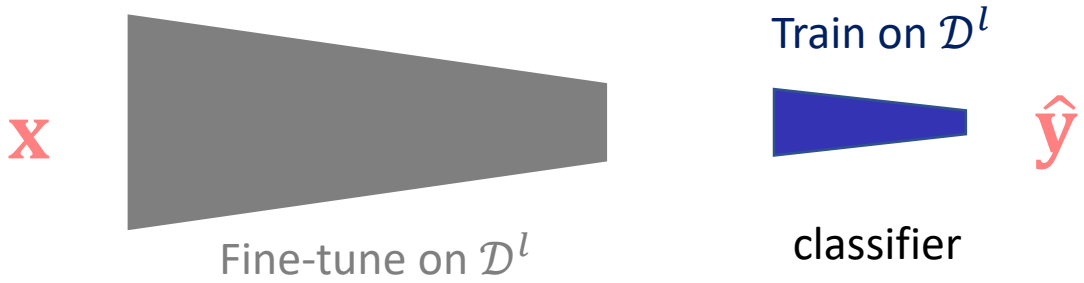
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Pre-training

- Train most parameters (e.g., early layers) using unlabelled data



- Train some small set of specific parameters (e.g., later layers) using labelled data



Through Generative Modeling

Classification using generative model

$$P(y|\mathbf{x},\boldsymbol{\theta}) = \frac{P(y|\boldsymbol{\theta})P(\mathbf{x}|y,\boldsymbol{\theta})}{P(\mathbf{x}|\boldsymbol{\theta})} = \frac{P(y|\boldsymbol{\theta})P(\mathbf{x}|y,\boldsymbol{\theta})}{\sum_{y\in\mathcal{Y}} P(y|\boldsymbol{\theta})P(\mathbf{x}\,|y,\boldsymbol{\theta})}$$

Maximum Likelihood estimation of generative parameters

$$\underset{\theta}{\operatorname{argmax}}\, P\left(\mathcal{D}^l, \mathcal{D}^u \middle| \boldsymbol{\theta}\right) = \prod_{i=1}^{n^l} P(\mathbf{x}_i^l, y_i^l\,|\boldsymbol{\theta}) \prod_{i=1}^{n^u} P(\mathbf{x}_i^u\,|\boldsymbol{\theta})$$

Conditional generative model

$$= \prod_{i=1}^{n^l} P(y_i^l|\boldsymbol{\theta})P(\mathbf{x}_i^l|y_i^l,\boldsymbol{\theta}) \prod_{i=1}^{n^u} P(\mathbf{x}_i^u\,|\boldsymbol{\theta})$$

Marginalizing class out as a latent variable

parametric prior on classes

$$= \prod_{i=1}^{n^l} P(y_i^l|\boldsymbol{\theta})P(\mathbf{x}_i^l|y_i^l,\boldsymbol{\theta}) \prod_{i=1}^{n^u} \sum_{y\in\mathcal{Y}} P(y|\boldsymbol{\theta})P(\mathbf{x}_i^u\,|y,\boldsymbol{\theta})$$

[Kingma et al., "Semi-supervised Learning with Deep Generative Models", NeurIPS 2014]

Through Generative Modeling

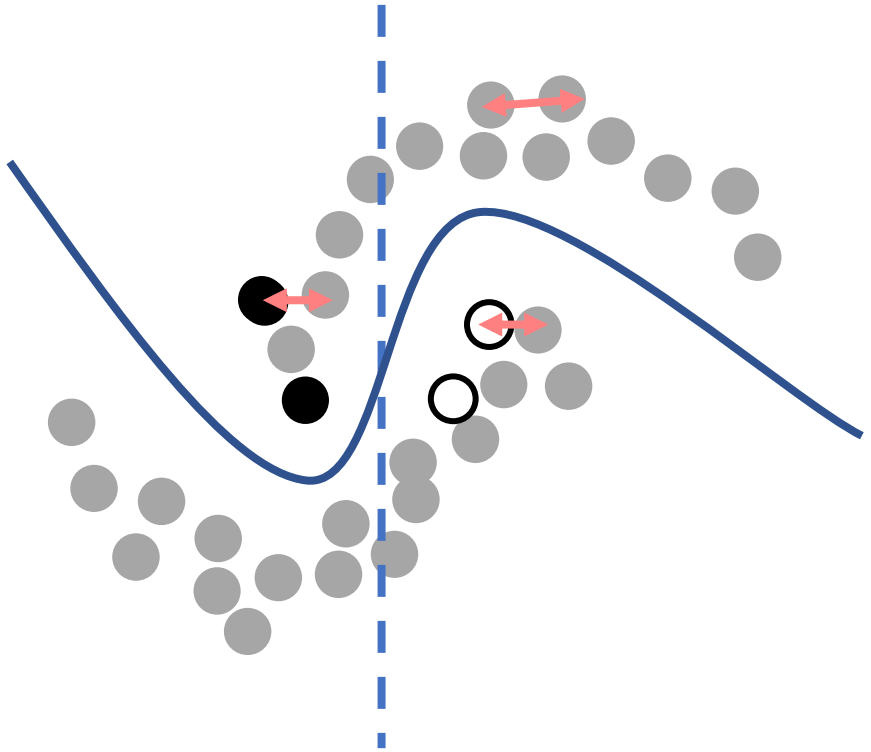
- Theoretically a sound way to incorporate unsupervised samples by probabilistic modeling
- many deep generative models are being developed
- Many more (generative) parameters than discriminative modeling needs
- Generative features can be irrelevant
- Misalignment of the parametric distributions



Semi-supervised Learning

Principle of Continuity/Smoothness/Manifold/Clustering/…

- **Pseudo labels**
- **Geodesic distance**
- **Clustering**
- **…**



Pseudo labels (also called self-training for single-view and co-training for multiple view)

Assume a network's predictions are calibrated

$\mathcal{D} = \mathcal{D}^l$

For some iterations:

train $\boldsymbol{\theta}^* = \text{argmin}_{\boldsymbol{\theta}} \mathcal{L}(\mathcal{D})$

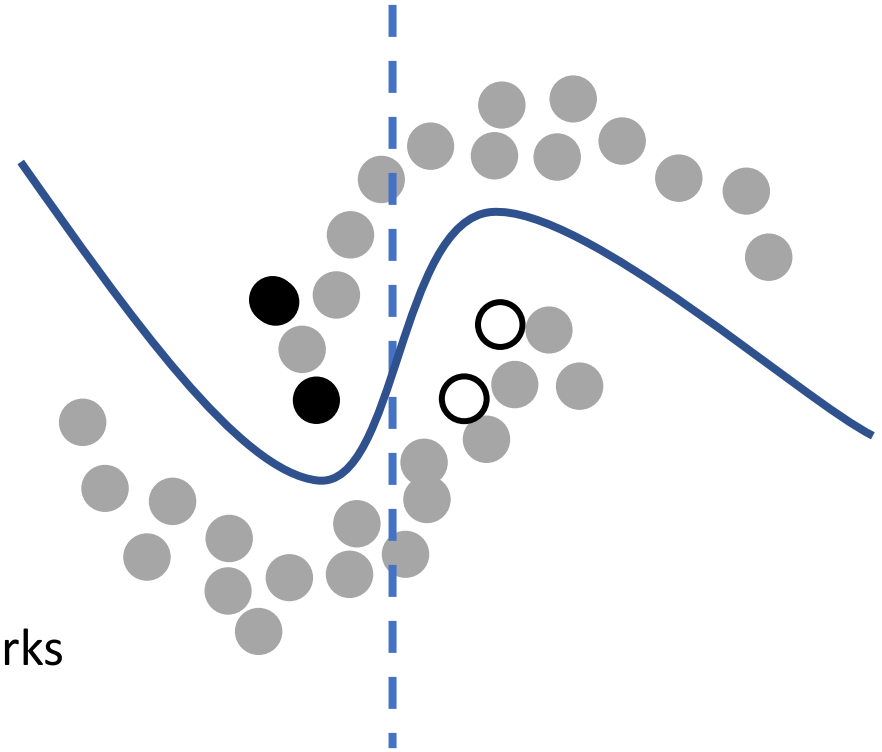
$\mathcal{D}^u_{conf} = \{\}$

for i = 1 : n^u

if $\text{conf}(f(x_i^u)) > threshold$

$\mathcal{D}^u_{conf} := \mathcal{D}^u_{conf} \cup x_i^u$

$\mathcal{D} = \mathcal{D}^l \cup \mathcal{D}^u_{conf}$



- You need a confidence measure
  - e.g., uncertainty estimation for probabilistic deep networks
- and a few other design choices

[Lee, "Pseudo-Label : The Simple and Efficient Semi-Supervised Learning Method for Deep Neural Networks", ICML workshop]

Pseudo Labelling

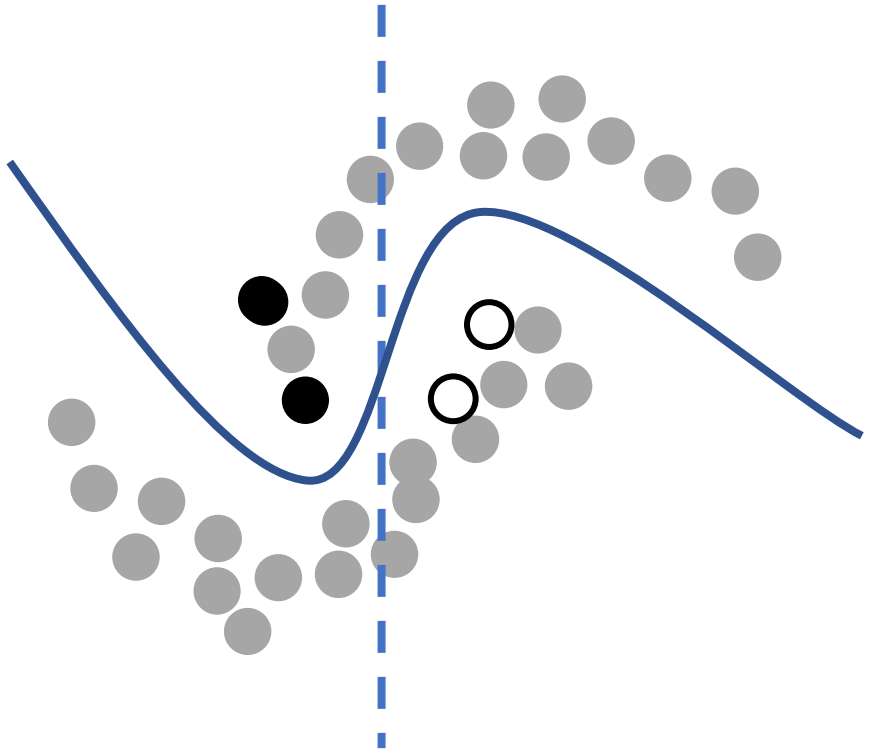
- Essentially the same as an idea from Yoshua Bengio from NIPS 2005!
- Y Grandvalet, Y Bengio, “Semi-supervised learning by entropy minimization”, NIPS 2005
- Explain this paper and talk about reinventing ideas!



Semi-supervised Learning

Pseudo labels (also called self-training for single-view and co-training for multiple view)

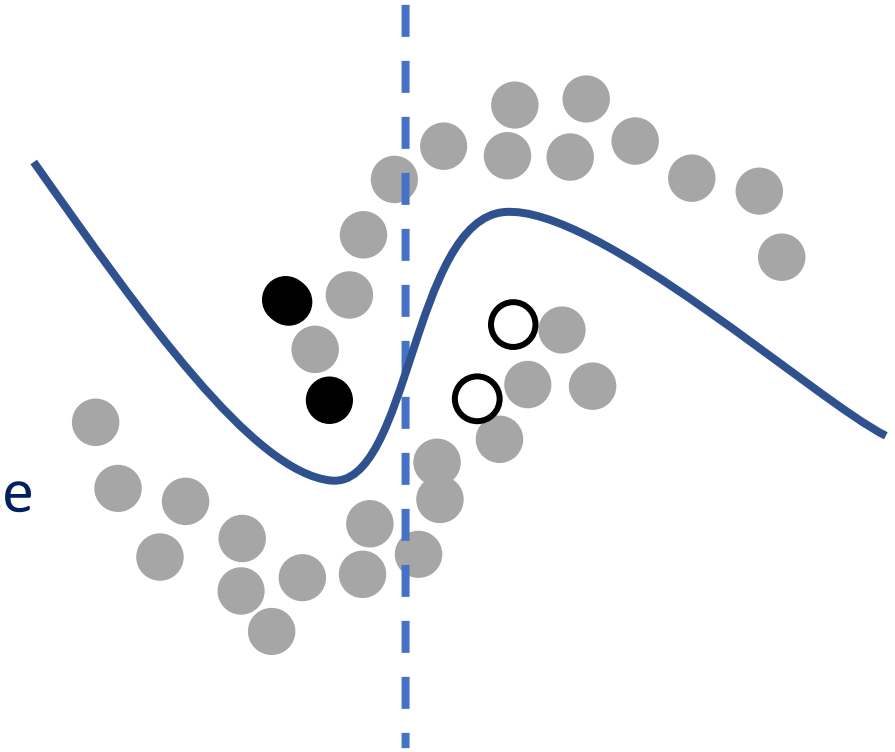
- Very simple to implement
- Applies to any base network
- Lingering (reinforced) misclassifications



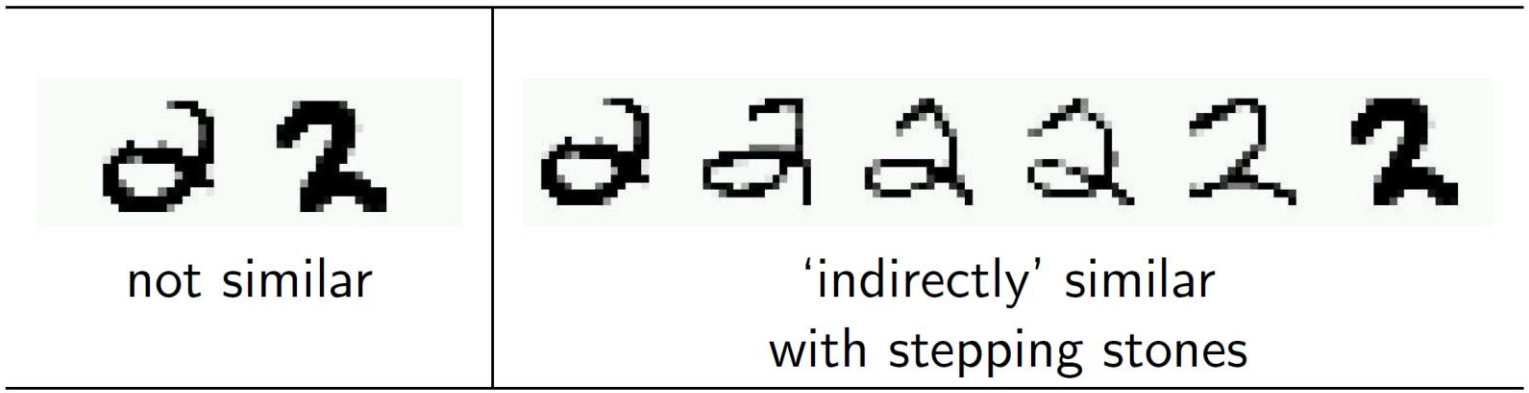
[Lee, “Pseudo-Label : The Simple and Efficient Semi-Supervised Learning Method for Deep Neural Networks”, ICML workshop]

Principle of Continuity – Geodesic distance

- $\mathcal{G} = (\mathcal{V}, \mathcal{E})$
- There is a node per data $|\mathcal{V}| = \left|\mathcal{D}^l\right| + |\mathcal{D}^u|$
- Take a distance measure in a proper feature space
  - $d_\varphi(\mathbf{x}_1, \mathbf{x}_2) = d\big(\varphi(\mathbf{x}_1), \varphi(\mathbf{x}_2)\big)$
- Construct a (similarity) graph based on the distance
  - $d_{i,j} = d\left(\varphi(\mathbf{x}_i), \varphi\big(\mathbf{x}_j\big)\right)$
  - sparse graph: $\mathcal{E} = \{(\mathbf{x}_i, \mathbf{x}_j) | d_{i,j} < t\}$
  - complete graph: $\mathcal{E} = \{(\mathbf{x}_i, \mathbf{x}_j, d_{i,j})\}$



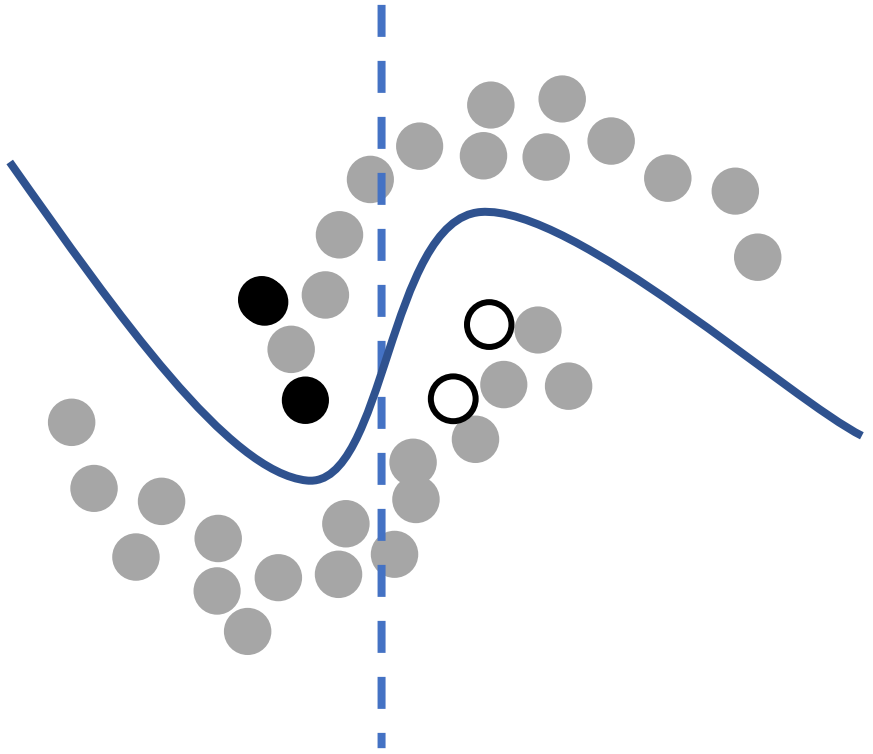
Principle of Continuity – Geodesic distance



[Example from ICML'07 tutorial by Xiaojin Zhu]

Principle of Continuity – Geodesic distance

- $\mathcal{G} = (\mathcal{V}, \mathcal{E})$
- $d_{\mathcal{G}}(\mathbf{x}_1, \mathbf{x}_2) = d_{\mathcal{G}}(\boldsymbol{v}_1, \boldsymbol{v}_2)$ (Graph Shortest Path)
- Several methods
  - Transfer the closest label
  - Mincut
  - …



[Fujiwara, Irie, "Efficient Label Propagation", ICML 2014]

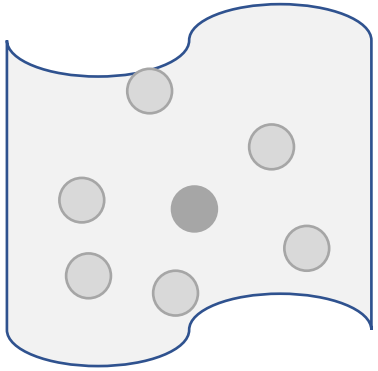
Semi-supervised Learning

- **Principle of Continuity – Geodesic distance**
- Graph theory algorithms for studying the data
- Quite sensitive to the feature space, the distance measure and the graph structure



Semi-supervised Learning

Principle of Consistency (manifold)



original sample

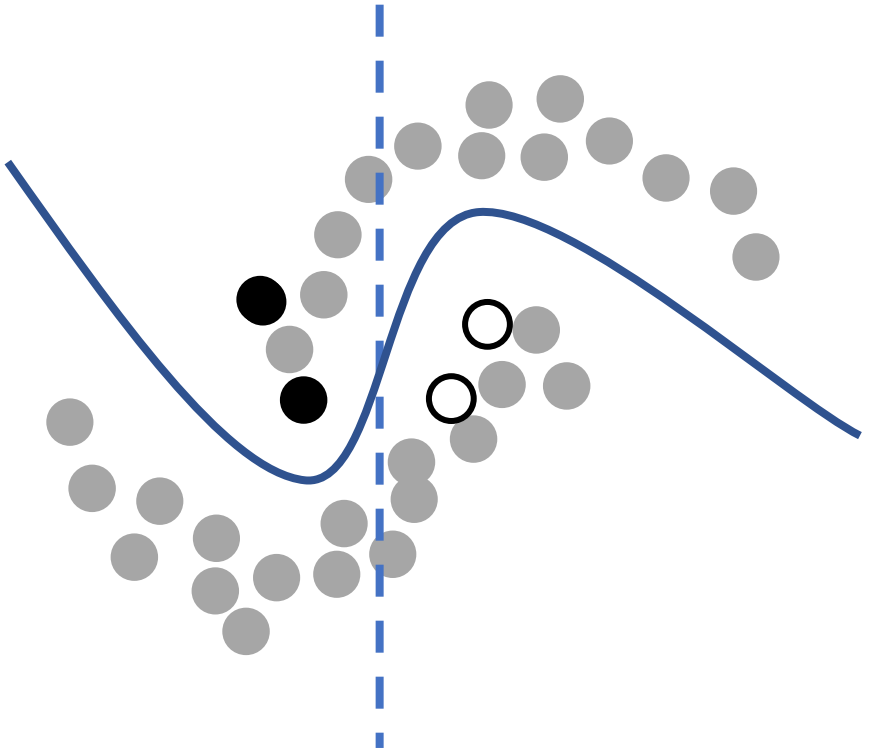
purturbed sample

data manifold

- $d\left(f_{\boldsymbol{\theta}}(\mathbf{x}), f_{\boldsymbol{\theta}}\left(perturb(\mathbf{x})\right)\right) < t$

- $\boldsymbol{\theta}^* = \text{argmin}_{\boldsymbol{\theta}}\, \mathcal{L}\left(\mathcal{D}^l\right) + \mathcal{L}^u\left(\mathcal{D}^l, \mathcal{D}^u\right)$

  $\mathcal{L}^u\left(\mathcal{D}^l, \mathcal{D}^u\right) = \sum_{i=1}^{n^u} d\left(f_{\boldsymbol{\theta}}(\mathbf{x}_i^u), f_{\boldsymbol{\theta}}\left(perturb(\mathbf{x}_i^u)\right)\right)$



Principle of Consistency – examples

- π-model
  - Perturbation
    - data augmentation
    - dropout
  - Distance
    - MSE: $\|f(\mathbf{x}) - f(perturb(\mathbf{x}))\|^2$

[Laine&Aila, "Temporal ensembling for semi-supervised learning", ICLR 2017]



Principle of Consistency – examples

- Virtual Adversarial Training (VAT)
  - Perturbation
    - adversarial example: $perturb(\mathbf{x}) = \mathbf{x} + \nabla_{\mathbf{x}}\, l(f(\mathbf{x}))$
  - Distance
    - KL-divergence

and there are many more cases of consistency-based semi-supervised learning in the literature

[Laine&Aila, “Temporal ensembling for semi-supervised learning”, ICLR 2017]



Notes

- it doesn't always work!
- most cases where it works satisfy the following conditions:
  - orders of magnitude more unlabelled samples than labelled ones
  - "too few" labelled samples w.r.t. the complexity of the task
  - a clearly overparameterized model w.r.t. the labelled dataset size
- for some domains it's more successful than others, e.g., NLP

